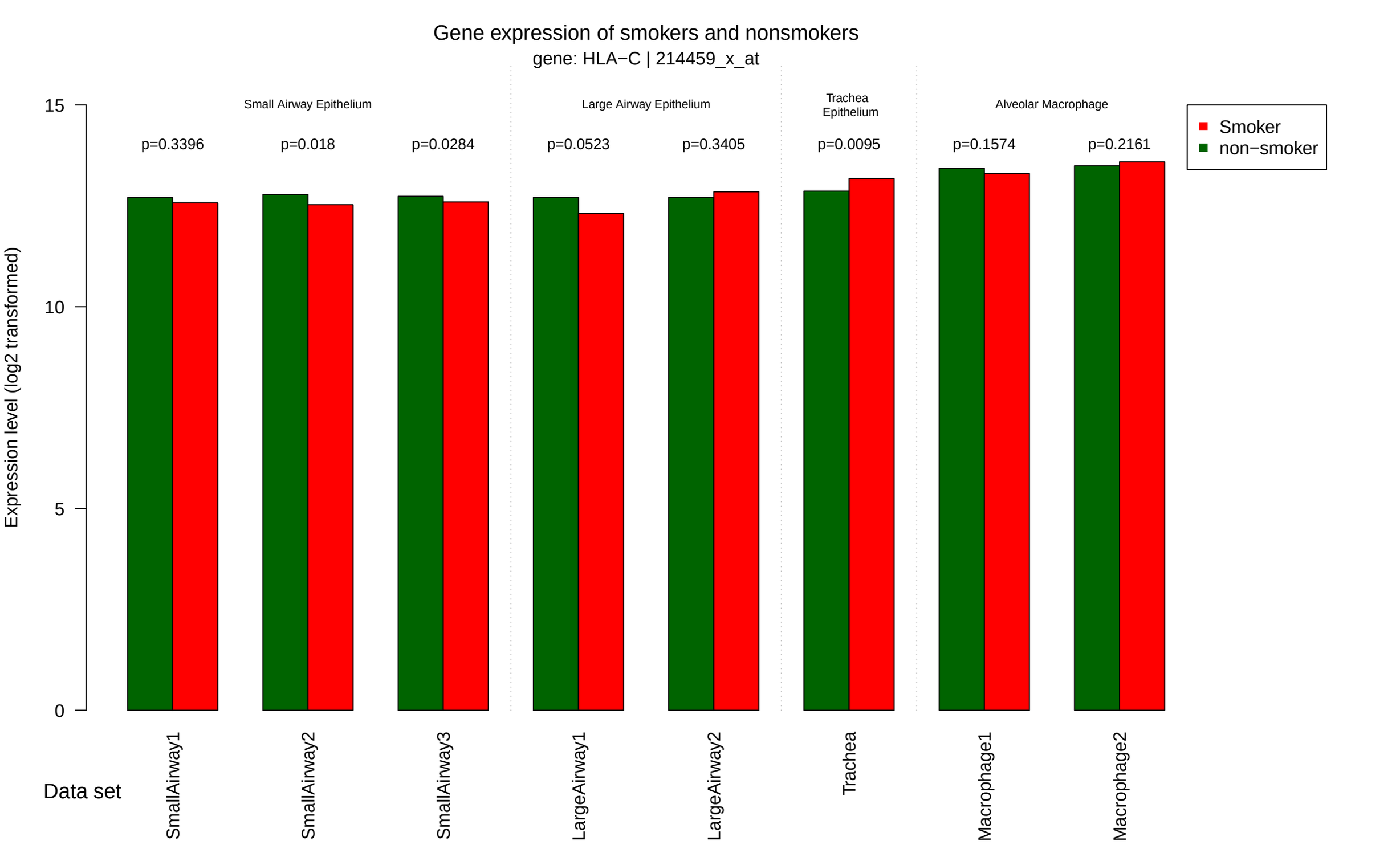
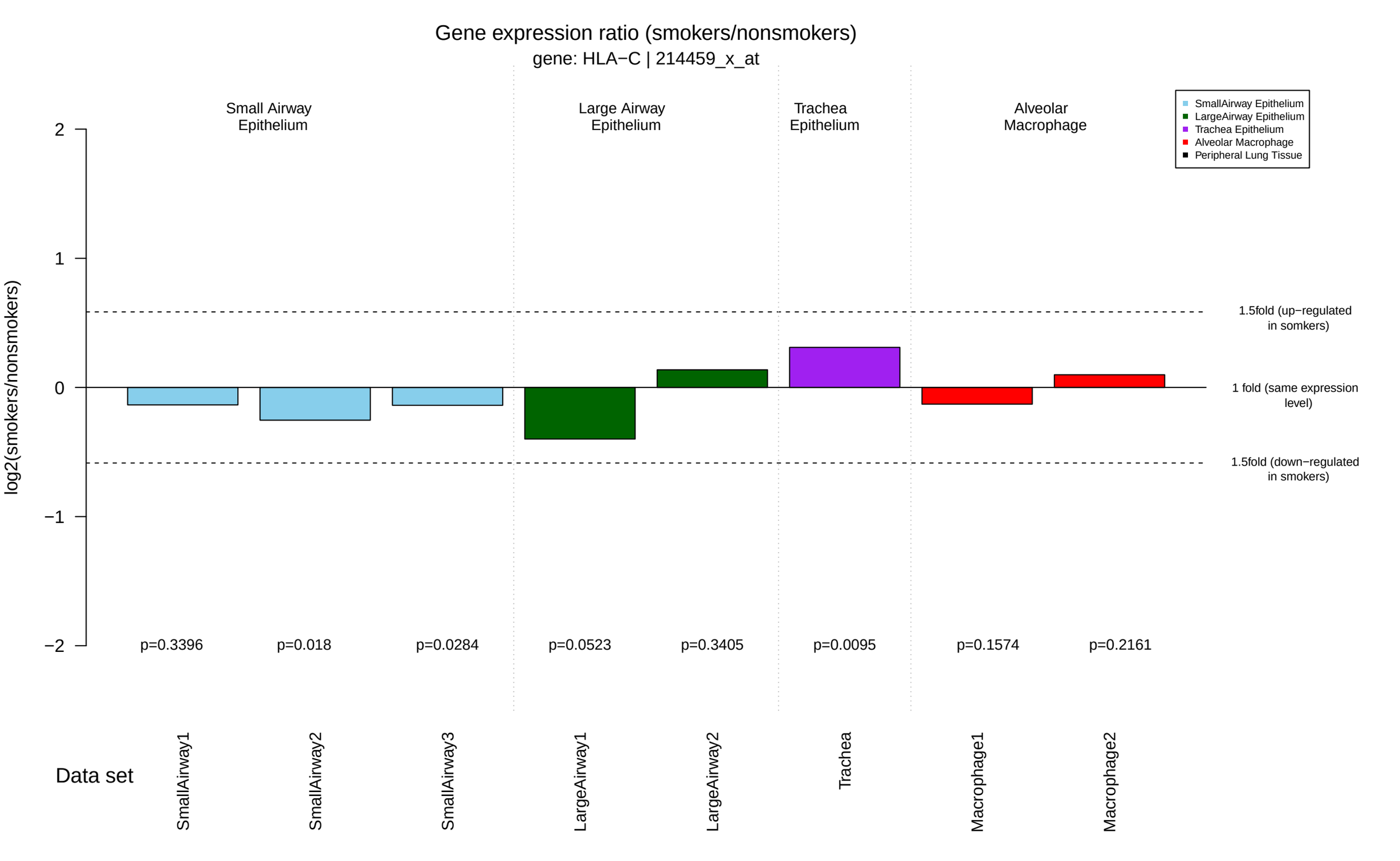
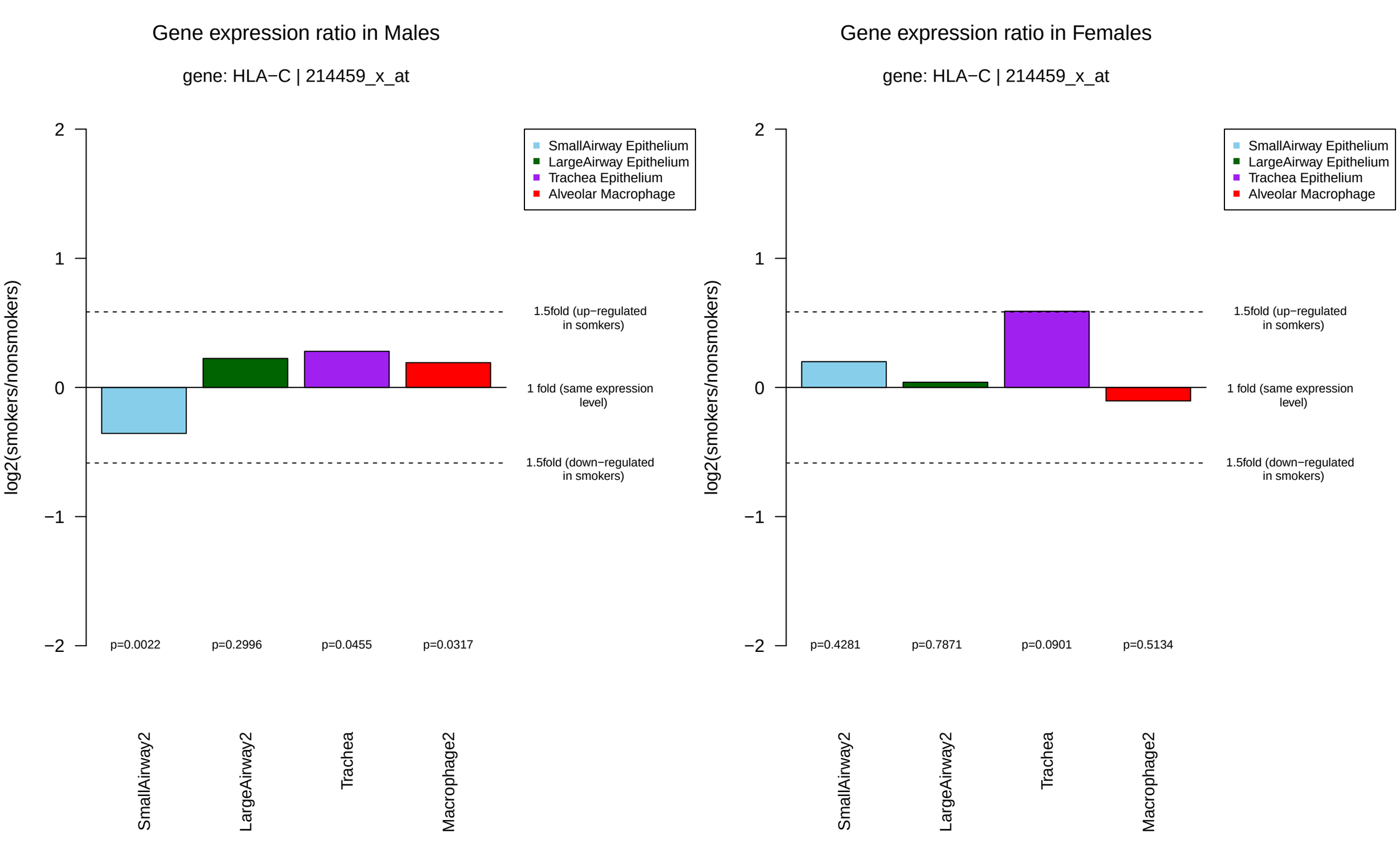
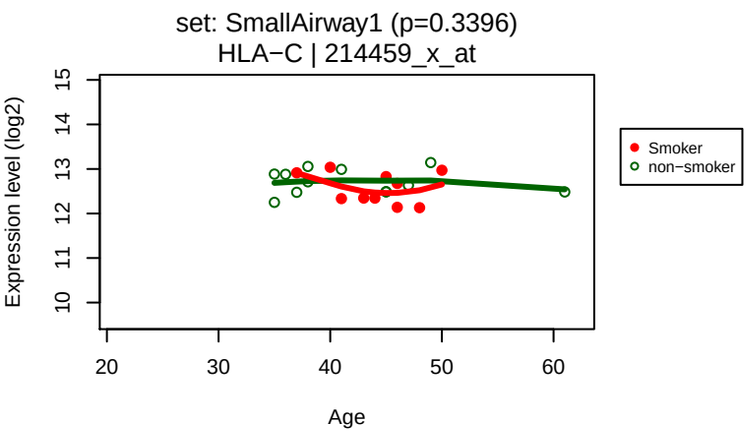
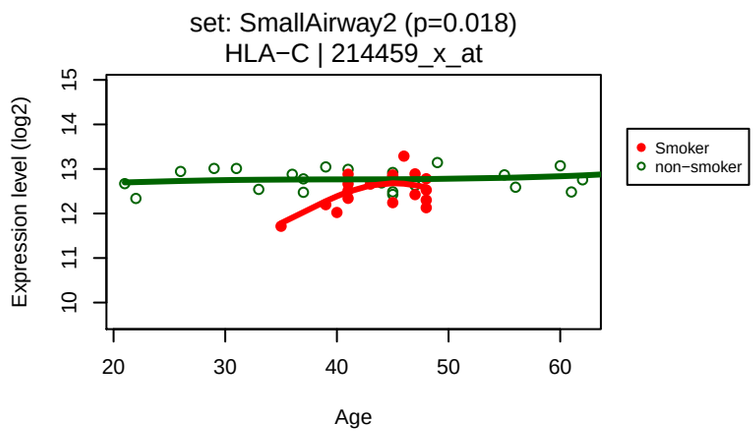
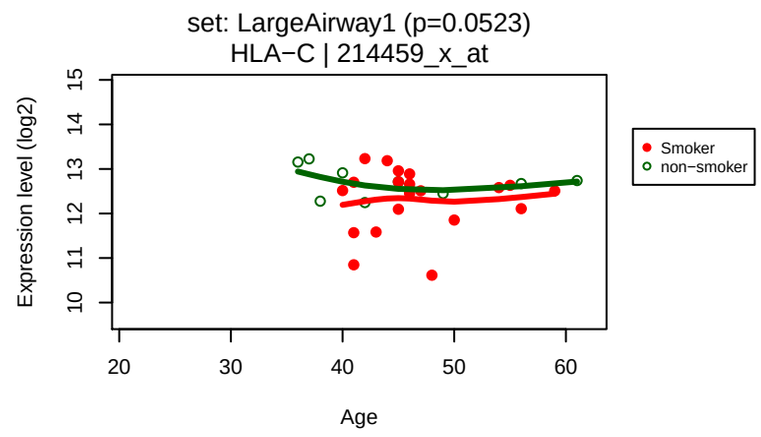

Gene expression of smokers and nonsmokers
gene: HLA−C | 214459_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3396
p=0.018
p=0.0284
p=0.0523
p=0.3405
p=0.0095
p=0.1574
p=0.2161
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

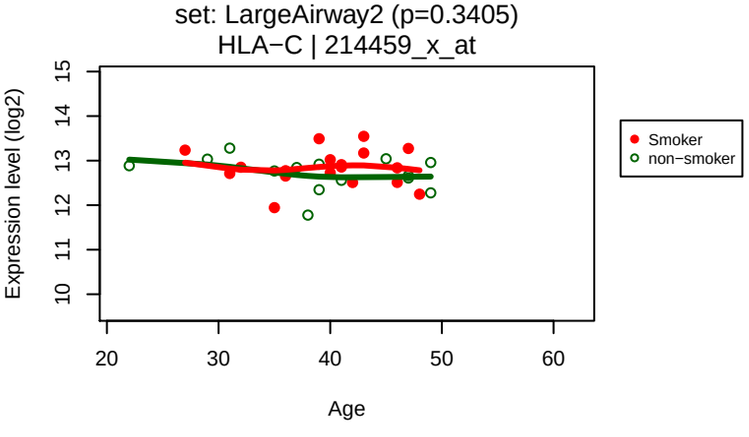

Gene expression ratio (smokers/nonsmokers)
gene: HLA−C | 214459_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3396
p=0.018
p=0.0284
p=0.0523
p=0.3405
p=0.0095
p=0.1574
p=0.2161
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

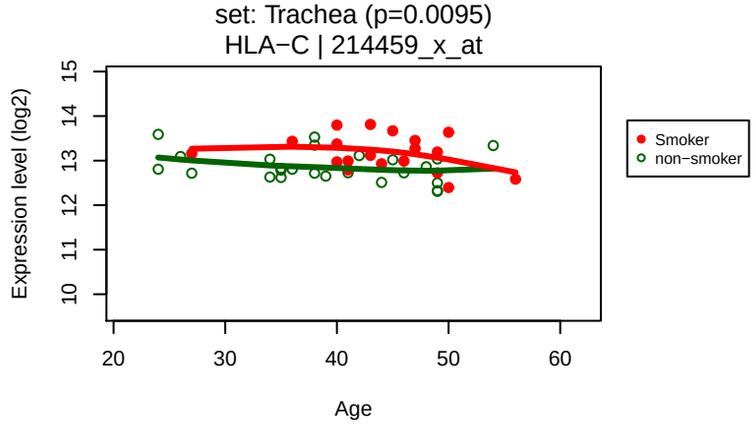

Gene expression ratio in Males
gene: HLA−C | 214459_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0022
p=0.2996
p=0.0455
p=0.0317
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−C | 214459_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4281
p=0.7871
p=0.0901
p=0.5134
SmallAirway2
LargeAirway2
Trachea
Macrophage2

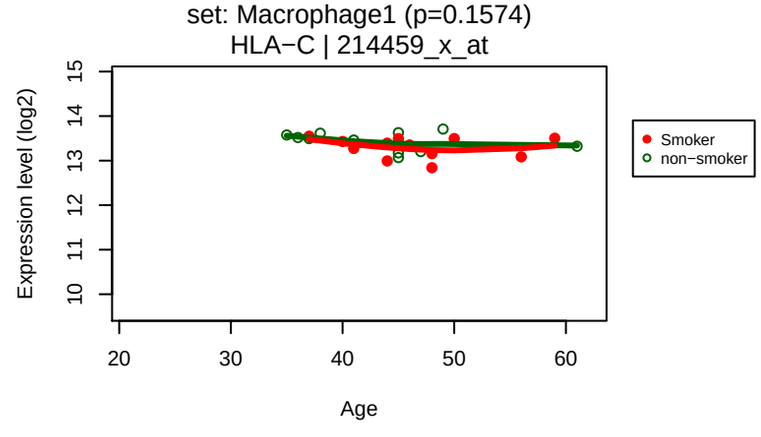

set: SmallAirway1 (p=0.3396)
HLA−C | 214459_x_at
Expression level (log2)
Age
Smoker
non−smoker

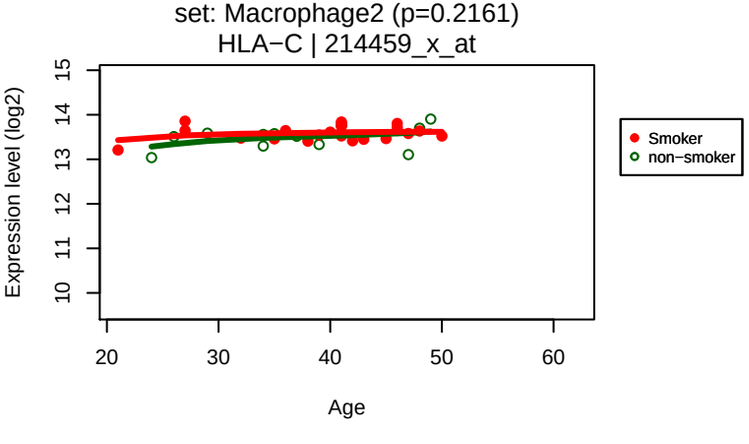

set: SmallAirway2 (p=0.018)
HLA−C | 214459_x_at
Expression level (log2)
Age
Smoker
non−smoker

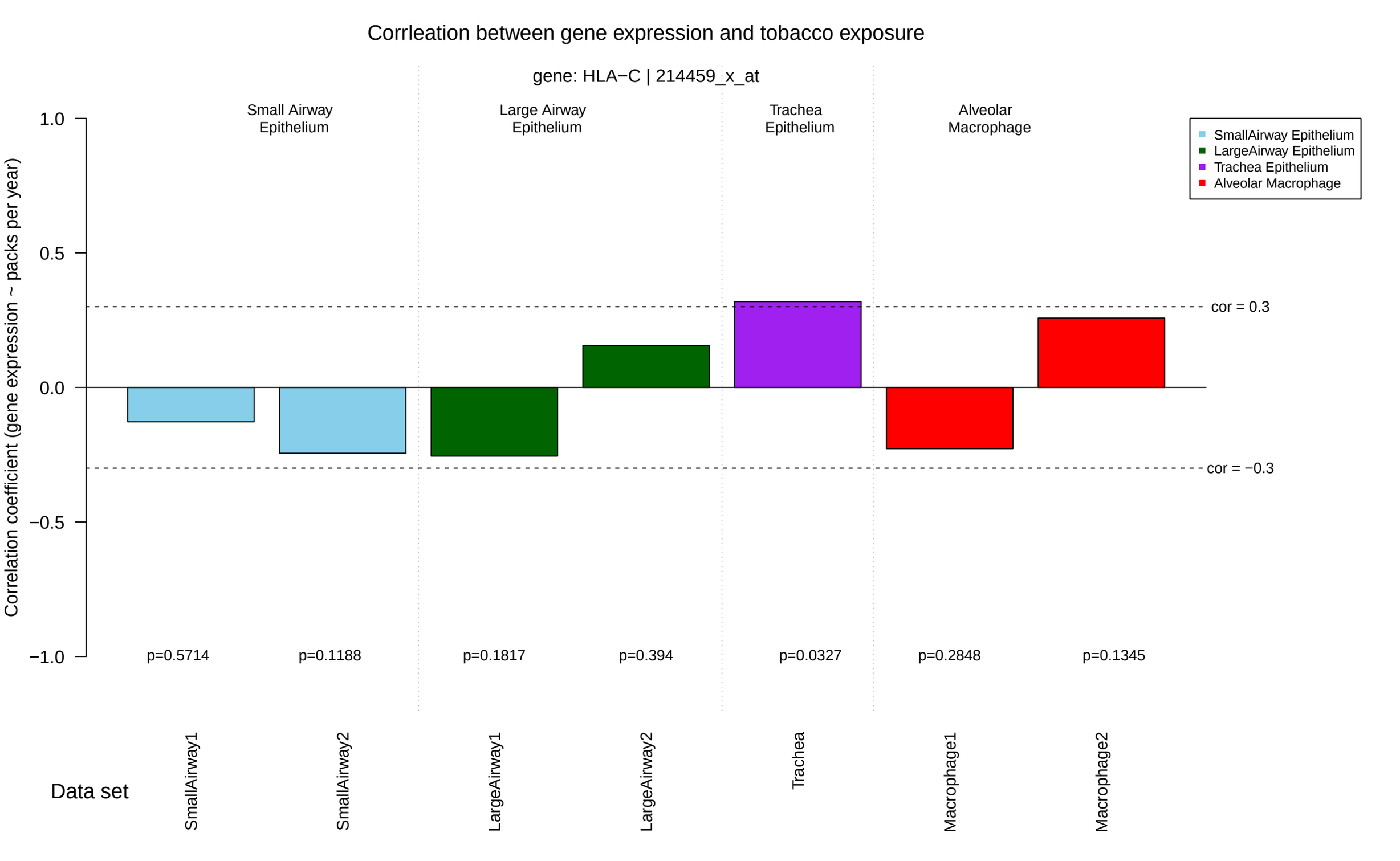

set: LargeAirway1 (p=0.0523)
HLA−C | 214459_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3405)
HLA−C | 214459_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0095)
HLA−C | 214459_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1574)
HLA−C | 214459_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2161)
HLA−C | 214459_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−C | 214459_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5714
p=0.1188
p=0.1817
p=0.394
p=0.0327
p=0.2848
p=0.1345
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2